CPSC 100

Computational Thinking

Data Mining

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Agenda

- Review clustering with KMeans and DBSCAN
- Cluster Quality
- File Systems

Course
Admin

Course Admin

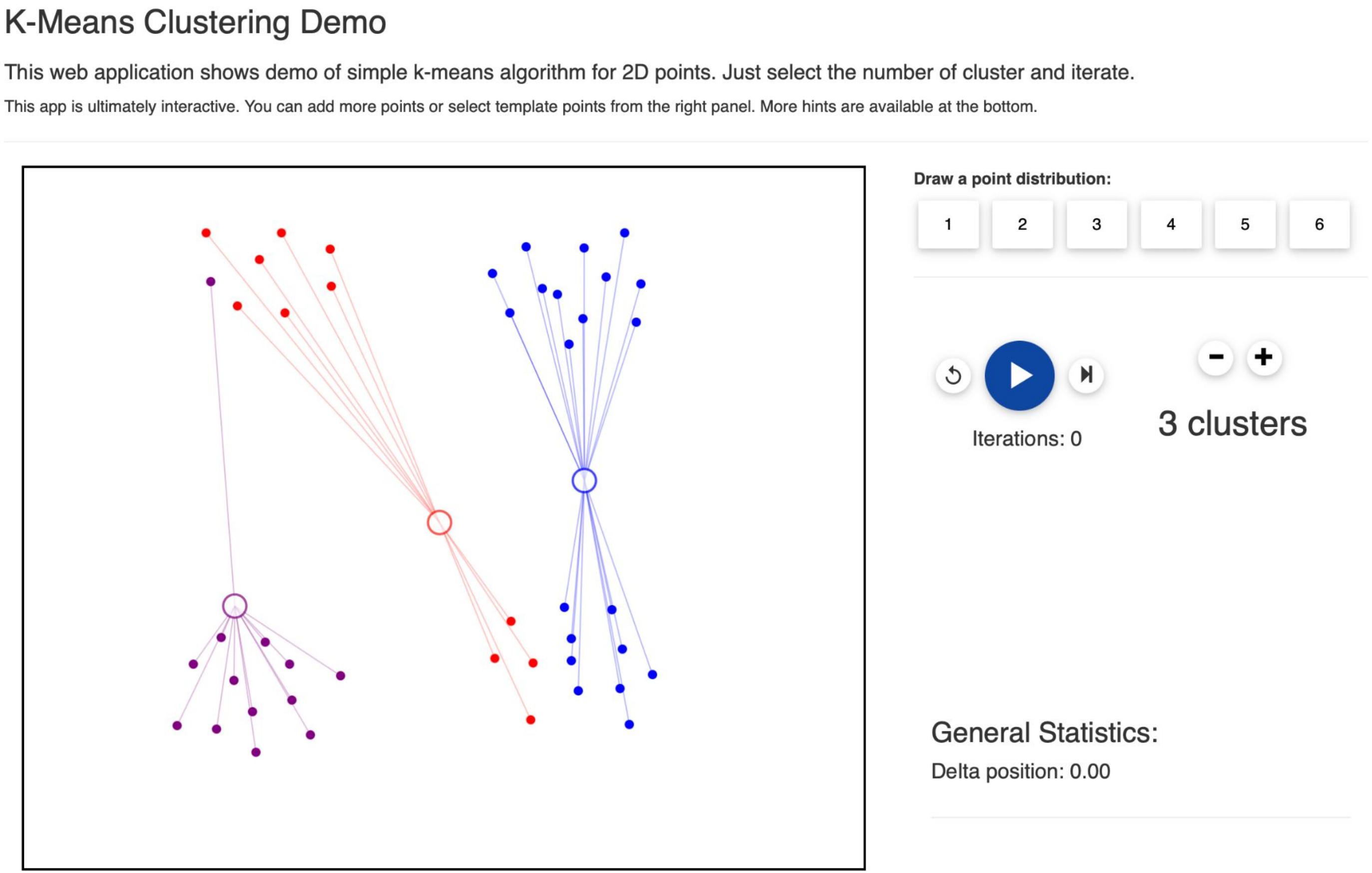
- Learning Log for this week to be released after class today (~5:30 PM) → Due Wednesday at 6 PM (Tuesday is a holiday)

- Remember: No classes and no labs next week!
  – TA Student hours are ON Thursday and Friday

- Goal is:
  – Project Milestone 2 feedback to you by Tuesday morning

  – Project Milestone 3 released by tomorrow morning

Another KMEANS Clustering Demo



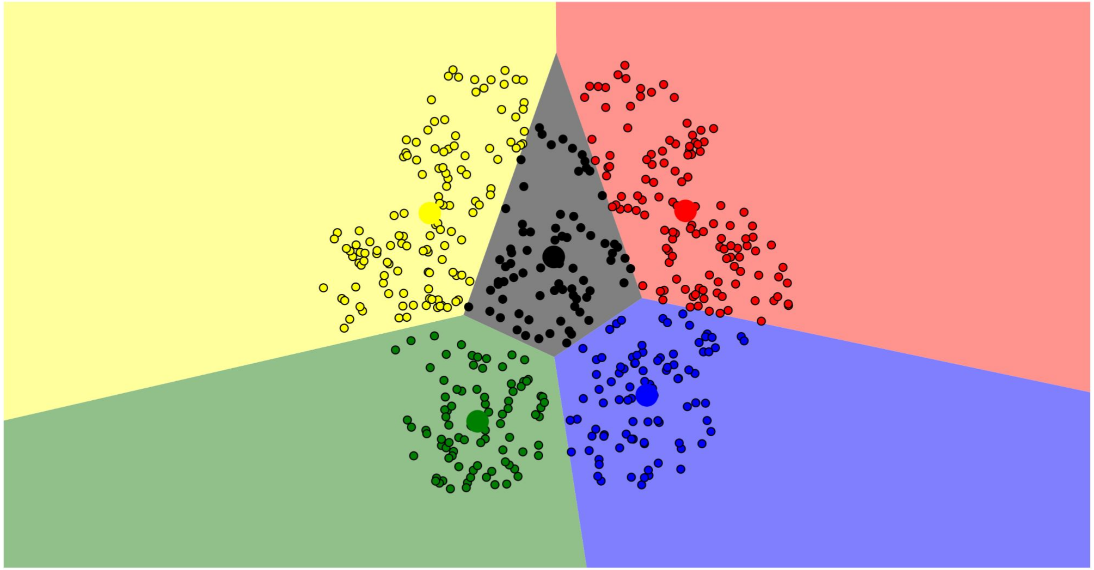
Source

Another KMEANS Clustering Demo

Visualizing K-Means Clustering

January 19, 2014

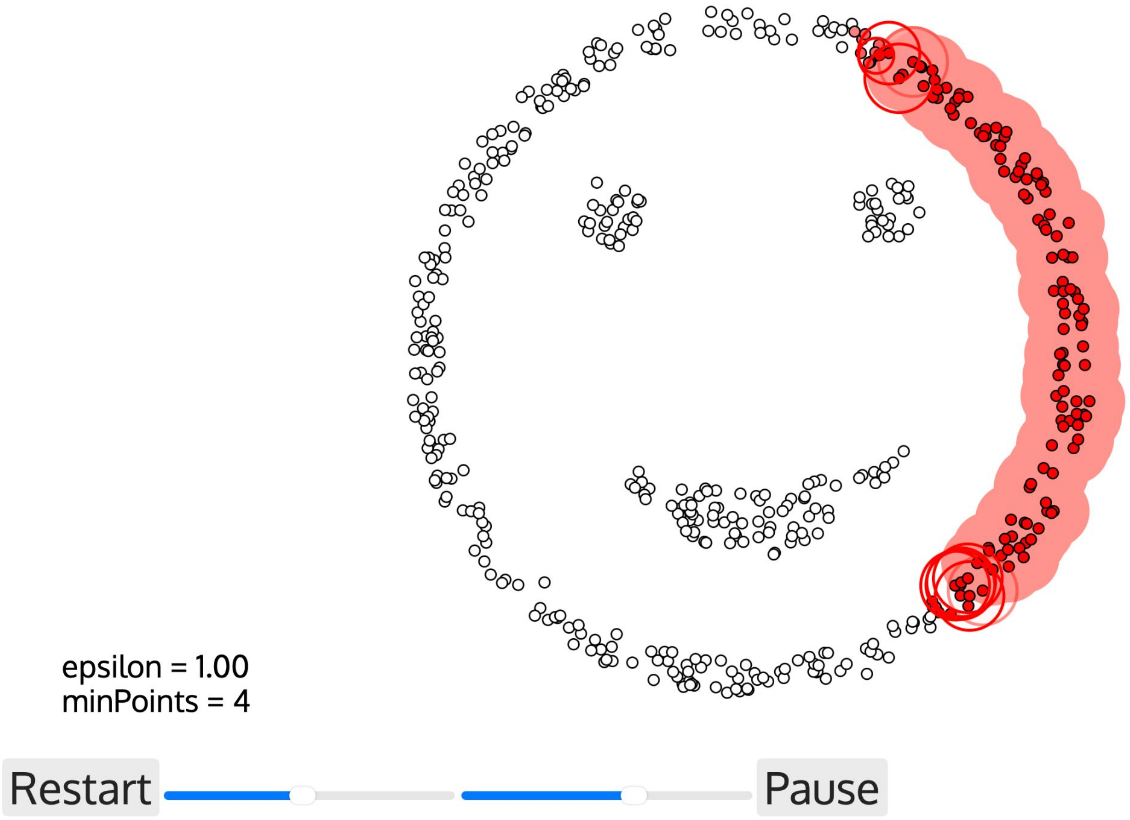
Suppose you plotted the screen width and height of all the devices accessing this website. You'd probably find that the points form three clumps: one clump with small dimensions, (smartphones), one with moderate dimensions, (tablets), and one with large dimensions, (laptops and desktops). Getting an algorithm to recognize these clumps of points without help is called *clustering*. To gain insight into how common clustering techniques work (and don't work), I've been making some visualizations that illustrate three fundamentally different approaches. This post, the first in this series of three, covers the k-means algorithm. To begin, click an initialization strategy below:



Restart Add Centroid Reassign Points

Source

Another DBSCAN Clustering Demo



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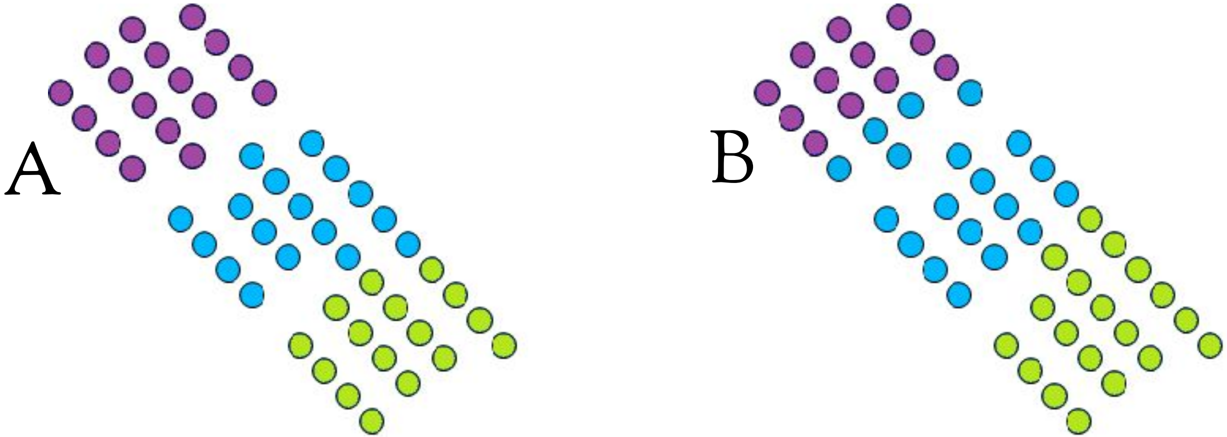
Apply the Knowledge

Cluster Quality

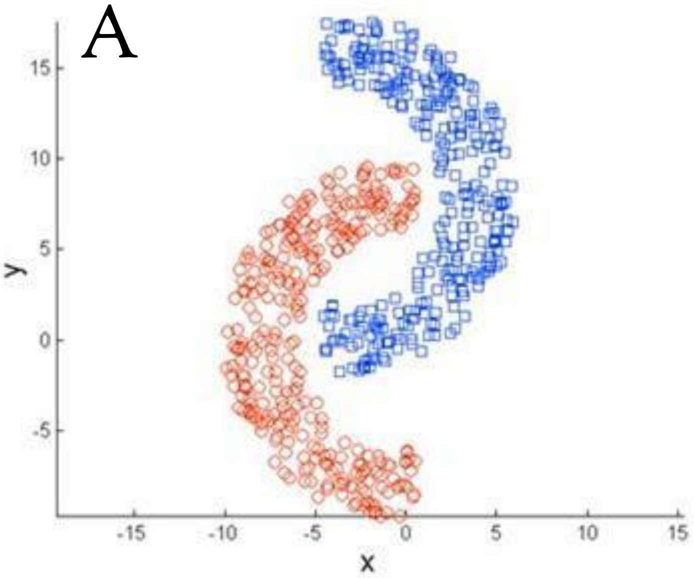
- Consider the following potential clusterings for assigning children to three different schools. Each point represents where a child lives.

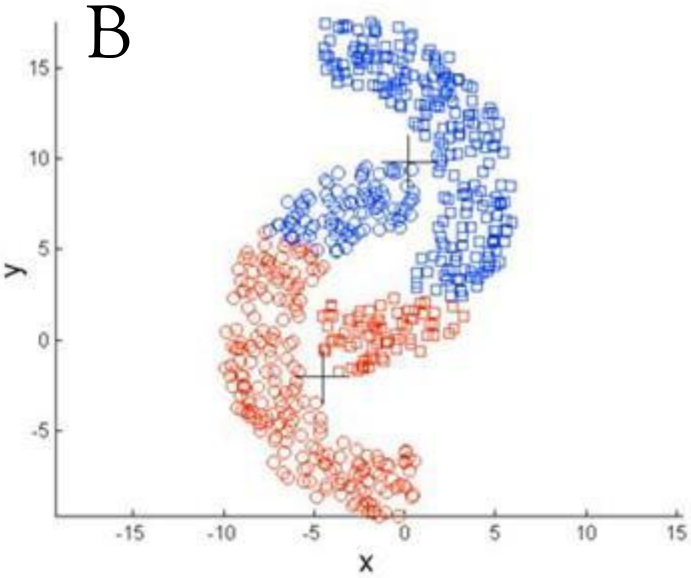
  What are the relative merits of each clustering?



Consider the following potential clusterings.

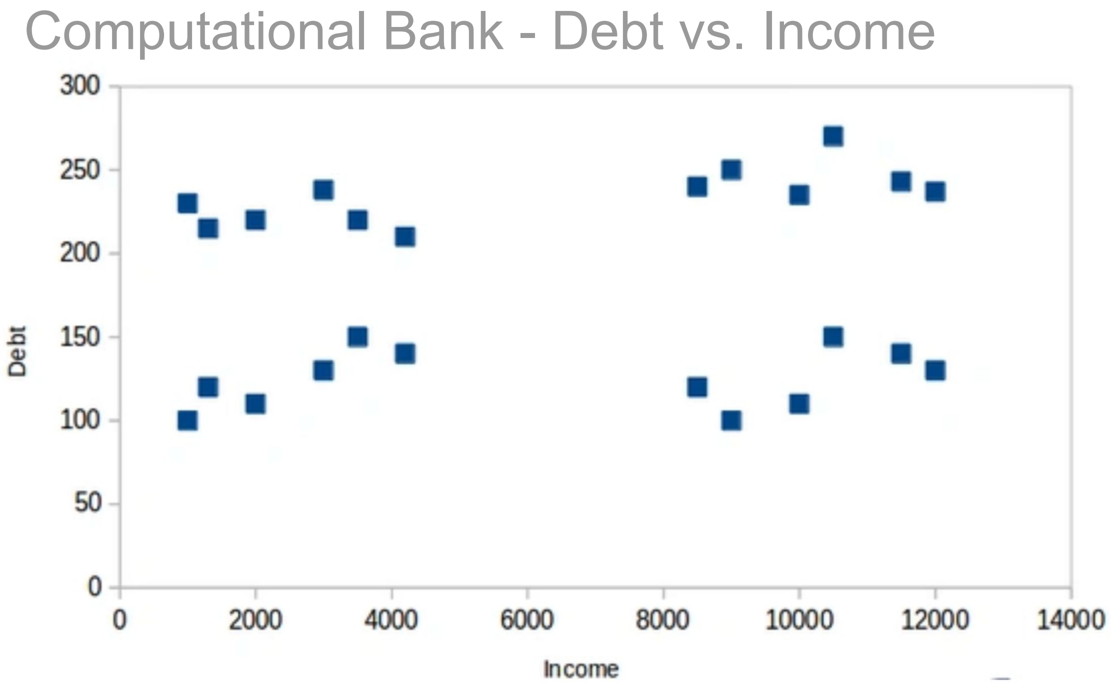
- What are the relative merits of each clustering?
- Which clustering was done by KMeans and which one by DBSCAN? How do you know?





Imagine you work for a bank and you only have data about the **income** (how much they earn) of your customers, and their **debt** (how much they owe).

Here's a plot of the data on an X-Y axis:



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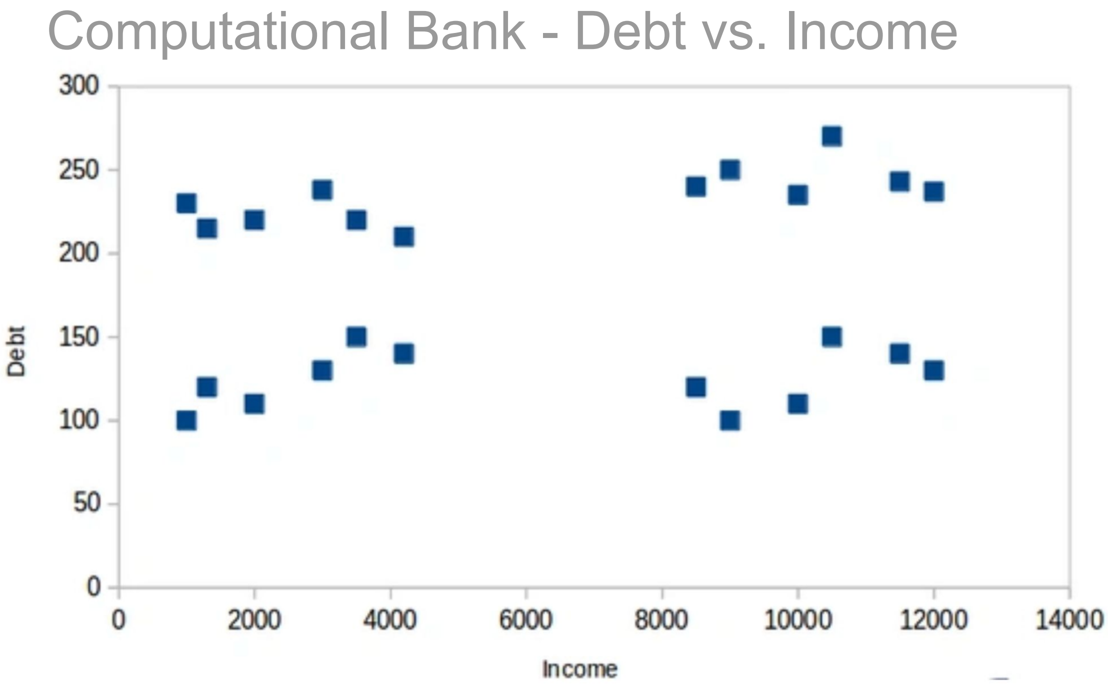
Imagine you work for a bank and you only have data about the **income** (how much they earn) of your customers, and their **debt** (how much they owe).

Cluster these data by eye - how many clusters would you choose? What would they be? Who is most valuable to the bank? Who should they target?



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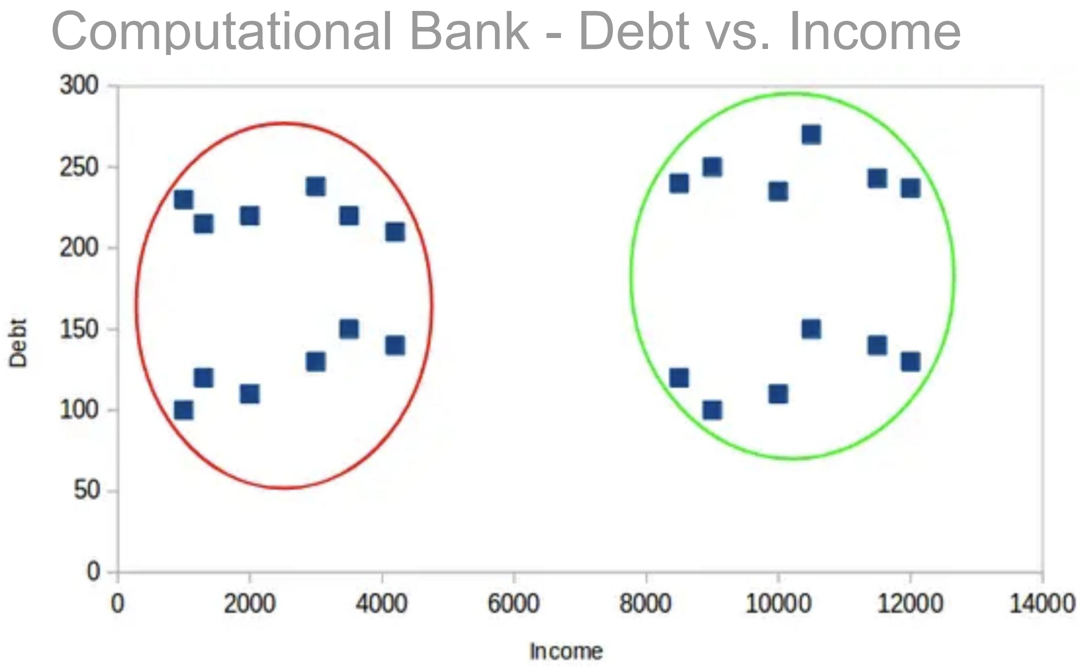

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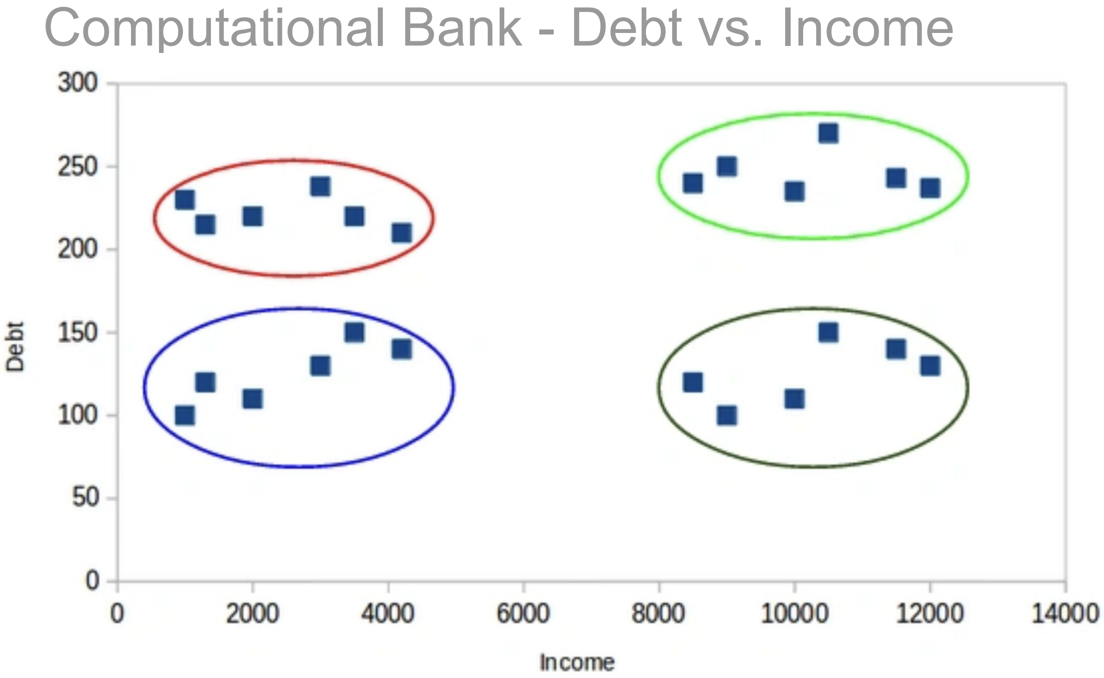

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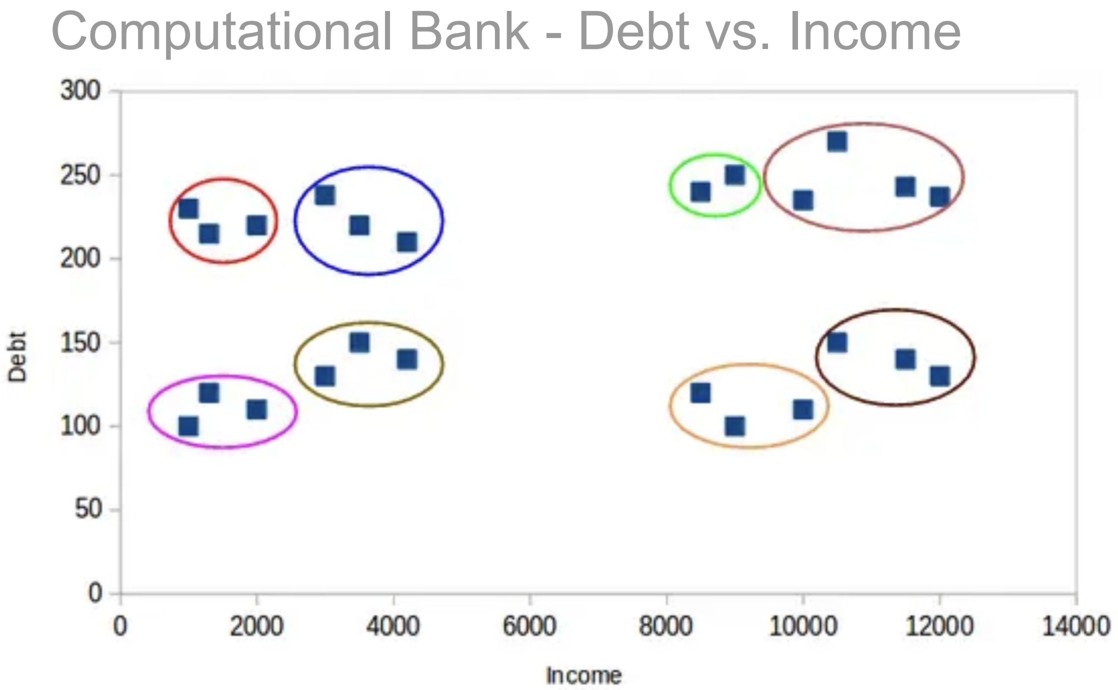


Source

File Systems

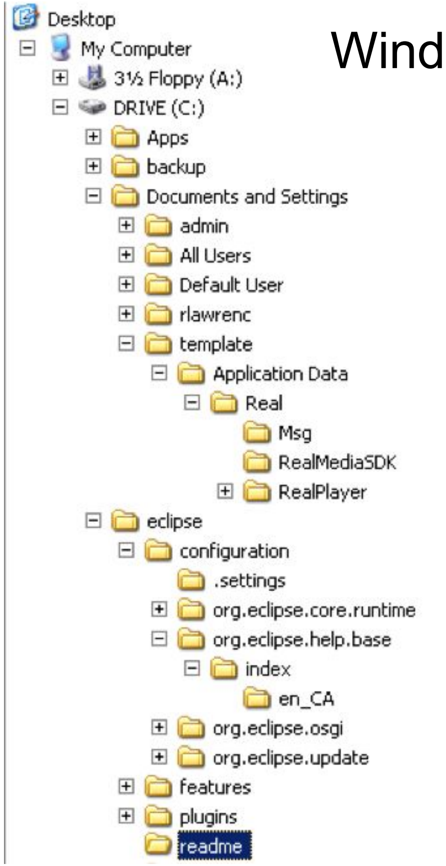
File Systems

The file system organizes data on a device as a hierarchy of **directories** and **files** (like a tree).

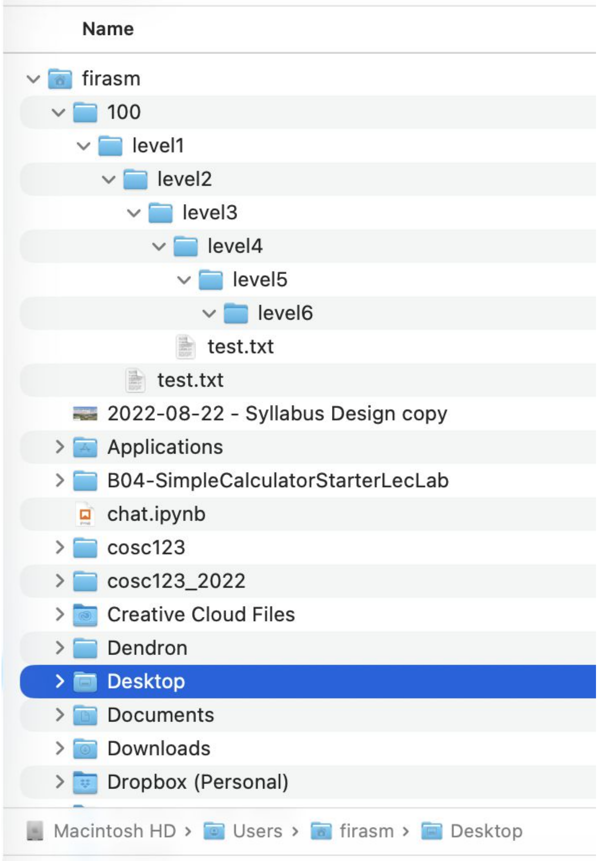
- Each folder (AKA directory) has a name and can contain any number of files or subdirectories.
- Each file has a name.
- The user can change (navigate) directories in the hierarchy.

File Systems

Windows

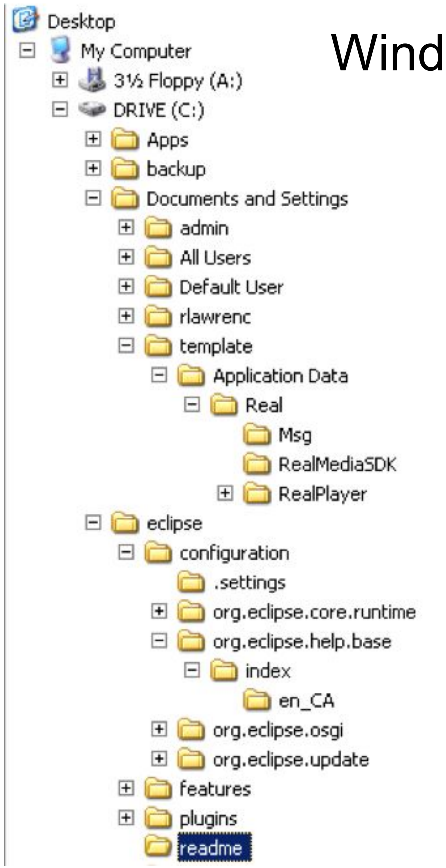


macOS



File Systems

Windows



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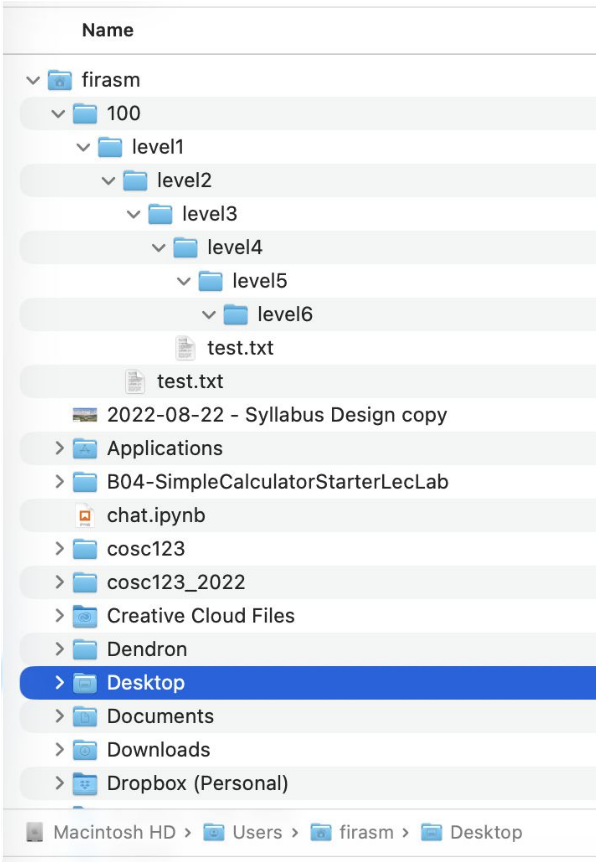


File Systems - Root (Children & Parents)

- The tree is rooted at, well, the root and there is **only one root** of a directory hierarchy.
- Every item in the tree is either a file or a directory (AKA folder).
- You can think of a directory as a container that may contains files and/or other directories.
- Files on the other hand hold information (and cannot contain other files or directories) .
- Example:

    If `directoryC` is contained in `directoryP`, then `directoryC` is a child of `directoryP` and `directoryP` is said to be the parent to `directoryC`.

- A directory may have many children, but can only have one parent.

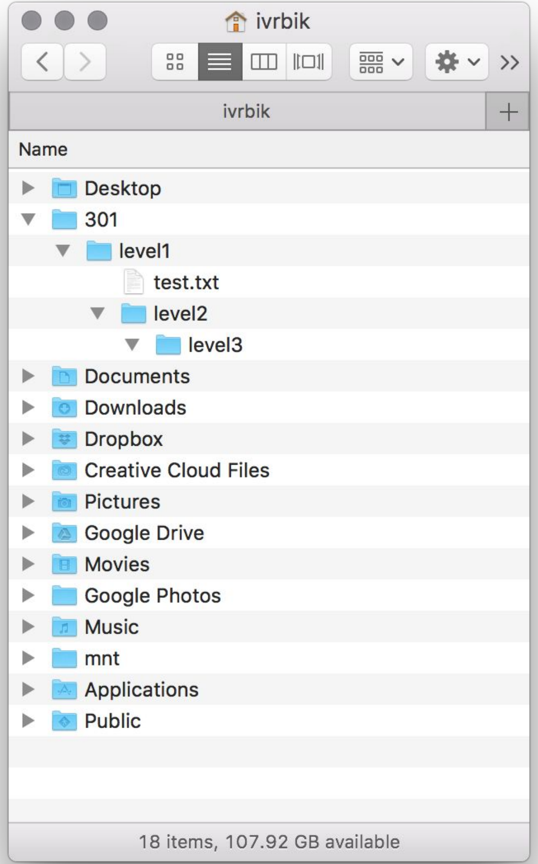
File Systems - Absolute and Relative Paths

- The root of the file system is the directory "/"

- There is only one root of a directory hierarchy.

- A path to a new location (from your current location) can be specified as an absolute path from the root (this will work no matter where we are in the file system):
  - . is the short-form for the current directory
  - .. signifies the parent directory (akin to pressing Cmnd + ↑ on a Mac, on Windows the shortcut is Alt+↑)
  - For example, to navigate (i.e. change directories) to the parent directory of the current directory, use the command: cd ..
- Note that this command is dependant on your current directory (i.e. the folder you are currently in).

File Systems - Absolute and Relative Paths



Absolute versus Relative Path Question

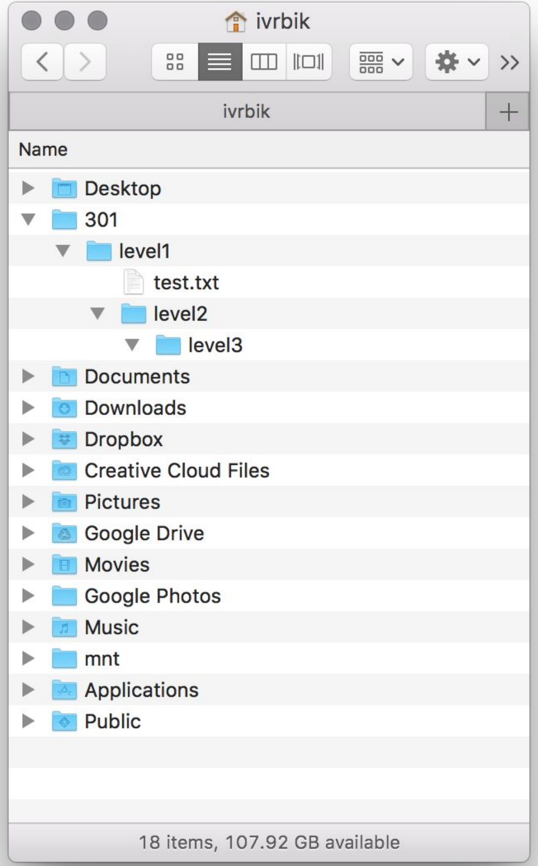
Consider this directory hierarchy.

- The user is currently in the directory `level2`
- The `level1` directory contains a file called `test.txt`.

Which of the following statements are TRUE?

1. A relative path to change to directory 301 is `..`
2. Absolute path to `test.txt` is `/Users/ivrbik/301/level1/test.txt`
3. Relative path to `test.txt` is ../test.txt
4. Relative path to `test.txt` is different if user was currently in `level3` directory
5. There is only one root of the directory hierarchy.

Activity



Try re-create the directory on the left and file structure in your HOME directory starting from:

On macOS:

/Users/ivrbik/**301** **---> onwards**

On Windows:

C:\Users\ivrbik\301**---> onwards**

Notice the slash directions!

Wrap up